

Telecom Customer Churn Analysis

An end-to-end data science project analyzing customer attrition patterns, identifying key churn drivers, quantifying revenue loss, and generating actionable retention strategies for a telecom company using Python.

DATA ANALYST

PYTHON

CHURN ANALYSIS

Executive Summary

Customer churn is one of the most critical challenges faced by telecom companies because it directly impacts revenue and profitability. This project analyzes customer churn patterns using Python to identify key factors influencing customer attrition and quantify the associated revenue loss.

Approach

The analysis involved data cleaning, transformation, exploratory data analysis, statistical testing, and business insight generation.

Variables Analyzed

Various customer attributes such as contract type, internet service type, tenure, offers, payment methods, and churn reasons were analyzed to understand their relationship with churn behavior.

Business Problem

The telecom company is experiencing customer churns, resulting in revenue loss. The objective of this analysis is to identify drivers, quantify losses, understand at-risk segments, and recommend retention strategies.

1

Identify Churn Drivers

Identify major drivers of customer churn.

2

Quantify Revenue Loss

Quantify revenue loss due to churn.

3

Segment Risk

Understand customer segments with higher churn risk.

4

Actionable Recommendations

Provide actionable recommendations to improve customer retention.

Dataset Overview

The dataset contains telecom customer information spanning demographics, service subscriptions, usage patterns, contract details, and revenue metrics — a comprehensive foundation for churn analysis.



Demographics

Customer demographics



Service Subscriptions

Service subscriptions



Internet Usage

Internet usage



Contract Information

Contract information



Payment Methods

Payment methods



Churn Reasons

Churn reasons



Revenue Metrics

Revenue metrics

Data Cleaning & Transformation

The following preprocessing steps were performed to prepare the dataset for rigorous exploratory and statistical analysis.

→ Column Cleanup

Removed unnecessary columns.

→ Missing Values

Identified and handled missing values.

→ Internet Nulls

Replaced internet-related null values for customers without internet service.

→ Churn Categories

Replaced missing churn categories and churn reasons for non-churned customers.

→ Binary Indicator

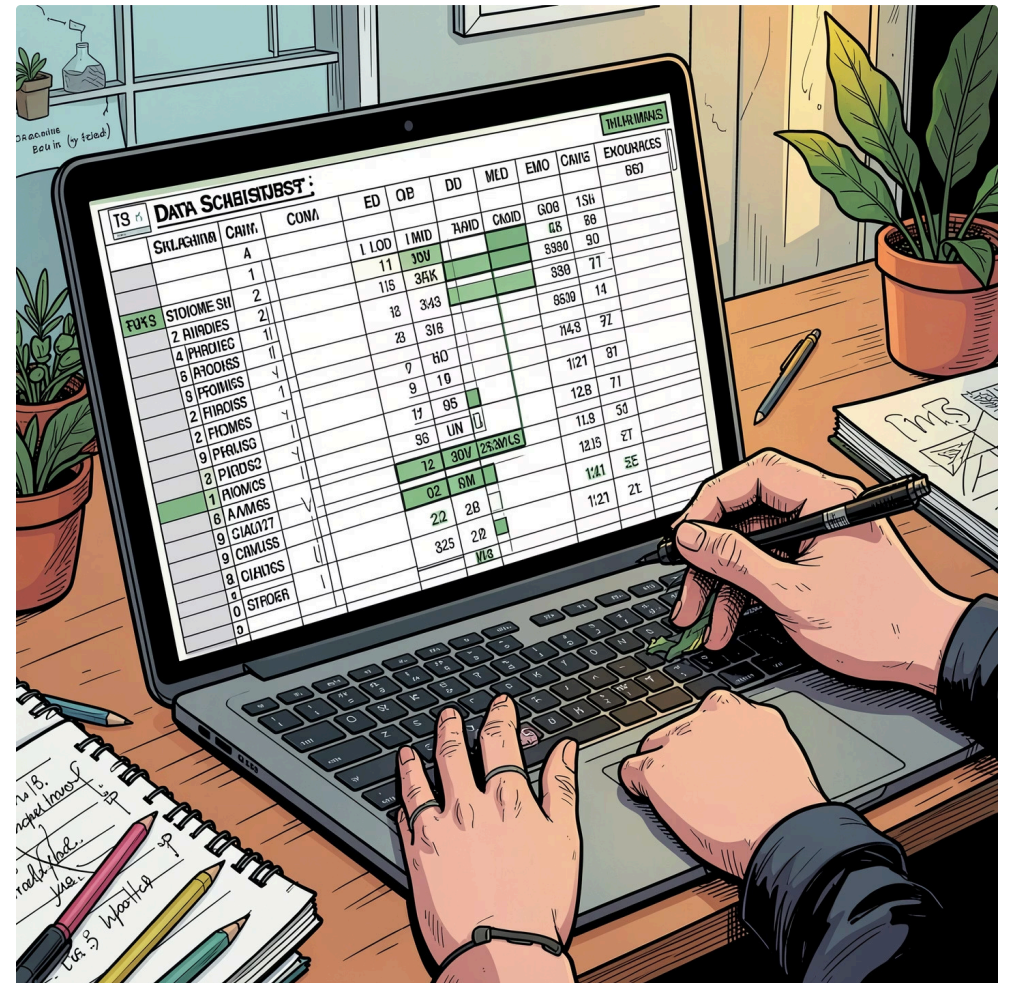
Created a binary churn indicator.

→ Age Groups

Created age group categories.

→ Tenure Categories

Created tenure categories.



Exploratory Data Analysis

Customer churn was analyzed across multiple dimensions to uncover patterns and correlations. Revenue loss was also examined across key churn categories.

Churn Dimensions Analyzed

- Gender
- Offer Type
- Internet Type
- Contract Type
- Payment Method
- Churn Category
- Churn Reason
- Customer Age Group
- Customer Tenure

Revenue Loss Analyzed Across

- Churn Categories
- Churn Reasons
- Contract Types
- Internet Types

Statistical Analysis

Rigorous statistical testing was applied to validate whether observed relationships between customer attributes and churn behavior were statistically significant.

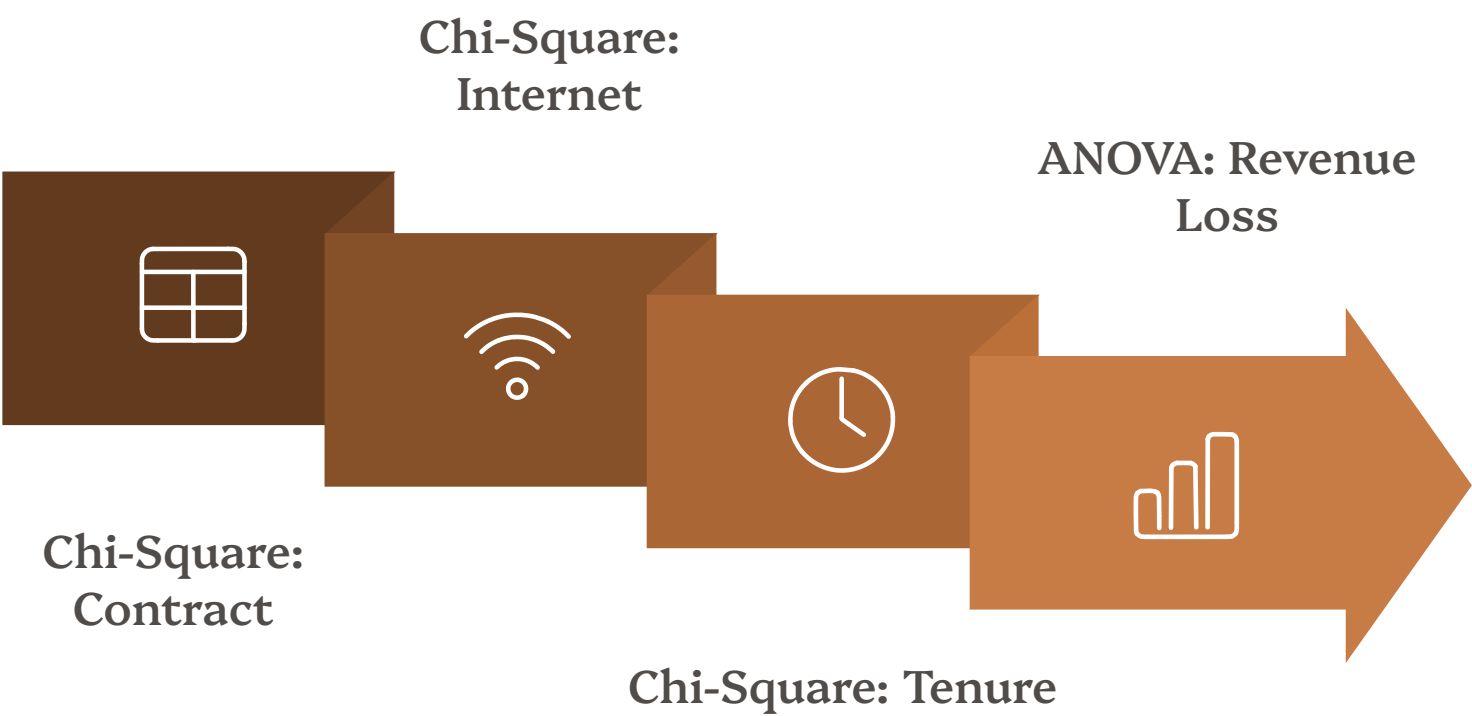
Chi-Square Tests

The following relationships were tested to determine whether customer churn is statistically associated with these variables:

- Contract Type vs Churn
- Internet Type vs Churn
- Tenure Category vs Churn

ANOVA Test

ANOVA was performed to determine whether revenue loss differs significantly across churn categories.



This two-phase approach first validated associations between categorical customer attributes and churn behavior, then assessed whether revenue loss varied significantly across churn categories.

Key Findings

Contract Structure

Customer churn is influenced by contract structure.

Internet Service Types

Certain internet service types exhibit higher churn rates.

Customer Tenure

Customer tenure is associated with churn likelihood.

Revenue Loss

Revenue loss varies across churn categories.

Competitor Reasons

Competitor-related reasons contribute significantly to churn.

Business Recommendations

Based on the analytical findings, the following targeted strategies are recommended to reduce churn and protect revenue.

1 Targeted Retention Campaigns

Develop targeted retention campaigns for high-risk customer segments.

2 Loyalty Benefits

Introduce loyalty benefits for long-tenure customers.

3 Competitive Positioning

Improve competitiveness against rival telecom providers.

4 Contract Review

Review contract offerings to reduce churn risk.

5 Proactive Monitoring

Monitor customer dissatisfaction indicators proactively.

Tools & Conclusion

Tools Used

Pandas	SciPy
Matplotlib	Seaborn
Jupyter	

Conclusion

This project demonstrates an end-to-end customer churn analysis workflow, including data preparation, exploratory analysis, statistical validation, and business insight generation. The findings can support data-driven customer retention strategies and revenue protection initiatives.

- ✅ The analytical framework and insights from this project can be directly applied to build proactive retention programs and reduce revenue loss from customer attrition.